

Operational & Financial Optimization in Patient Care

Data-driven insights from healthcare administrative & billing records

54,966 RECORDS

15 ATTRIBUTES

7-SLIDE REVIEW

THE MANAGEMENT QUESTION

How can we reduce
operational
friction while
aligning revenue
with care
complexity?

Executive analytics deck • Administrative, diagnostic and billing patterns

From raw admissions to analysis-ready records

A transparent cleaning and feature-engineering pipeline

CLEANED DATASET

54,966

patient admission
records

15 ATTRIBUTES

Demographics • diagnosis
billing • provider • dates

CLEANING PIPELINE

01

REMOVE DUPLICATES

534 duplicate rows removed

02

CORRECT BILLING

108 negative entries normalized

03

ENGINEER FEATURES

LOS • daily billing • age groups

WHY IT MATTERS

Accuracy

Prevents duplicate volume and revenue inflation.

Consistency

Normalizes anomalous billing values.

Usability

Creates comparable operational measures.

Four patterns management should investigate

The dataset shows strong uniformity where differentiation might be expected

33.1%

Inconclusive tests

18,198 cases returned an inconclusive result.

\$25.15K

Cancer avg billing

Near the same level as lower-complexity categories.

15.58 d

Emergency LOS

Elective admissions average 15.51 days.

\$215

Payer spread

Approximate range from \$25,415 to \$25,630.

BILLING BY CONDITION

Cancer		\$25,155
Diabetes		\$25,662
Obesity		\$25,807

READ-THROUGH

Billing appears highly compressed across conditions, warranting a cost-to-care review.

Two friction points: diagnostic velocity + pricing

Operational delay and financial uniformity are the central signals in this dataset

33.11%

DIAGNOSTIC INCONCLUSIVENESS

18,198 cases

returned an inconclusive outcome.

Redemption

Additional diagnosis codes should be required.

FLAT RATE

BILLING DIFFERENTIATION

Cancer

\$25,154.73

Diabetes

\$25,662.15

Obesity

\$25,806.63

MANAGEMENT IMPLICATION

Acuity-adjusted pricing should be evaluated only after internal resource and cost data are integrated.

Three moves from insight to execution

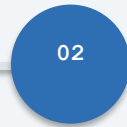
Prioritize diagnostic reliability, cost transparency, and discharge flow



LAB RELIABILITY

Recalibrate equipment
Update testing guidelines
Mandate secondary reviews

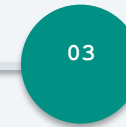
OWNER • CMO + Lab Director



ACUITY-BASED PRICING

Run ABC costing audit
Map resource consumption
Pilot before scale

OWNER • CFO + Revenue Cycle



ELECTIVE DISCHARGE

Dedicated discharge team
Standardize planning
Track bed turnaround

OWNER • COO + Nursing

Sequencing principle: fix data quality → establish true cost-to-care → then scale pricing and throughput changes.

Turn recommendations into measurable management

targets below are proposed operating goals, not observed outcomes



SUCCESS LOOP

MEASURE → DIAGNOSE → PILOT → VALIDATE → SCALE

Use the findings as a decision-support starting point

The current dataset supports pattern detection—not final financial or clinical conclusions

WHAT THE DATA CANNOT YET ANSWER

True profitability

Billed charges are not internal operating costs or profit.

Clinical acuity

No CPT/ICD severity controls are available.

Real-world validation

Synthetic uniformity requires comparison with EHR logs.

NEXT ANALYTICAL ROADMAP

01

Integrate cost accounting

Calculate departmental cost and margin.

02

Merge EHR clinical logs

Link treatment intensity to stay duration.

03

Pilot in one service line

Validate acuity-based billing before scale.

FINAL TAKEAWAY

Validate first. Pilot second. Scale with evidence.